

```
In [ ]: from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
In [ ]: # convert jupyter notebook to html file
! jupyter nbconvert --to html /content/DSCI303_Final_supervised_model_compar
```

```
In [ ]: import pandas as pd
import numpy as np

path = "/content/drive/MyDrive/Junior Spring/DSCI303/Final Project/Final_Aug
df = pd.read_csv(path)

df.head()
```

```
Out [ ]:
```

	diseases	anxiety and nervousness	depression	shortness of breath	depressive or psychotic symptoms	sharp chest pain	dizziness	inson
0	panic disorder	1	0	1	1	0	0	
1	panic disorder	0	0	1	1	0	1	
2	panic disorder	1	1	1	1	0	1	
3	panic disorder	1	0	0	1	0	1	
4	panic disorder	1	1	0	0	0	0	

5 rows x 378 columns

```
In [ ]: df.shape
```

```
Out [ ]: (246945, 378)
```

```
In [ ]: print("Shape:", df.shape)
print("Number of unique diseases:", df["diseases"].nunique())
print("\nFirst 10 columns:")
print(df.columns[:10].tolist())

print("\nHead:")
print(df.head())

print("\nMissing values:")
print(df.isnull().sum().sum())
```

Shape: (246945, 378)

Number of unique diseases: 773

First 10 columns:

```
['diseases', 'anxiety and nervousness', 'depression', 'shortness of breath',  
'depressive or psychotic symptoms', 'sharp chest pain', 'dizziness', 'insomnia',  
'abnormal involuntary movements', 'chest tightness']
```

Head:

	diseases	anxiety and nervousness	depression	shortness of breath	
\					
0	panic disorder	1	0		1
1	panic disorder	0	0		1
2	panic disorder	1	1		1
3	panic disorder	1	0		0
4	panic disorder	1	1		0
	depressive or psychotic symptoms		sharp chest pain	dizziness	insomnia
\					
0		1	0	0	0
1		1	0	1	1
2		1	0	1	1
3		1	0	1	1
4		0	0	0	1
	abnormal involuntary movements		chest tightness	...	\
0		0	1	...	
1		0	0	...	
2		0	0	...	
3		1	0	...	
4		1	1	...	
	stuttering or stammering		problems with orgasm	nose deformity	\
0		0	0	0	
1		0	0	0	
2		0	0	0	
3		0	0	0	
4		0	0	0	
	lump over jaw	sore in nose	hip weakness	back swelling	\
0	0	0	0	0	
1	0	0	0	0	
2	0	0	0	0	
3	0	0	0	0	
4	0	0	0	0	
	ankle stiffness or tightness		ankle weakness	neck weakness	
0		0	0	0	
1		0	0	0	
2		0	0	0	
3		0	0	0	
4		0	0	0	

[5 rows x 378 columns]

Missing values:

0

```
In [ ]: import matplotlib.pyplot as plt
# -----
# 2. Basic overview
# -----
print("Shape of dataset:", df.shape)
print("\nFirst 5 rows:")
print(df.head())

print("\nColumn names:")
print(df.columns.tolist())

print("\nData types:")
print(df.dtypes)

# -----
# 3. Identify label column
# -----
if "diseases" in df.columns:
    label_col = "diseases"
elif "Disease" in df.columns:
    label_col = "Disease"
else:
    label_col = df.columns[-1] # assume last column is target

print("\nLabel column:", label_col)

# -----
# 4. Missing values
# -----
missing_per_col = df.isnull().sum()
total_missing = missing_per_col.sum()

print("\nTotal missing values in dataset:", total_missing)
print("\nColumns with missing values:")
print(missing_per_col[missing_per_col > 0].sort_values(ascending=False))

# -----
# 5. Duplicate rows
# -----
num_duplicates = df.duplicated().sum()
print("\nNumber of duplicate rows:", num_duplicates)

# -----
# 6. Target variable analysis
# -----
print("\nNumber of unique diseases:", df[label_col].nunique())

class_counts = df[label_col].value_counts()

print("\nTop 10 most common diseases:")
print(class_counts.head(10))

print("\nTop 10 least common diseases:")
```

```

print(class_counts.tail(10))

print("\nSummary of class counts:")
print(class_counts.describe())

# -----
# 7. Feature analysis
# -----
X = df.drop(columns=[label_col])

print("\nNumber of feature columns:", X.shape[1])

# number of unique values in each feature
unique_counts = X.nunique()

print("\nFeature unique-value summary:")
print(unique_counts.describe())

# check if features are mostly binary
non_binary_cols = [col for col in X.columns if not set(X[col].dropna().unique())

print("\nNumber of non-binary feature columns:", len(non_binary_cols))
if len(non_binary_cols) > 0:
    print("Some non-binary columns:")
    print(non_binary_cols[:20])

# symptom frequency
symptom_counts = X.sum().sort_values(ascending=False)

print("\nTop 20 most frequent symptoms:")
print(symptom_counts.head(20))

print("\nTop 20 least frequent symptoms:")
print(symptom_counts.tail(20))

# -----
# 8. Rare classes
# -----
rare_classes = class_counts[class_counts < 2]
print("\nNumber of diseases with fewer than 2 samples:", len(rare_classes))

# optional
print("\nDiseases with fewer than 2 samples:")
print(rare_classes)

# -----
# 9. Quick plots
# -----

# class distribution
plt.figure(figsize=(10, 5))
class_counts.plot(kind="hist", bins=50)
plt.title("Distribution of Number of Samples per Disease")
plt.xlabel("Samples per Disease")
plt.ylabel("Number of Diseases")
plt.show()

```

```

# top 20 diseases
plt.figure(figsize=(12, 6))
class_counts.head(20).plot(kind="bar")
plt.title("Top 20 Most Common Diseases")
plt.xlabel("Disease")
plt.ylabel("Count")
plt.xticks(rotation=90)
plt.show()

# top 20 symptoms
plt.figure(figsize=(12, 6))
symptom_counts.head(20).plot(kind="bar")
plt.title("Top 20 Most Frequent Symptoms")
plt.xlabel("Symptom")
plt.ylabel("Count")
plt.xticks(rotation=90)
plt.show()

# -----
# 10. Useful summary prints
# -----
print("\nFinal EDA Summary")
print("-----")
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])
print("Features:", X.shape[1])
print("Unique diseases:", df[label_col].nunique())
print("Total missing values:", total_missing)
print("Duplicate rows:", num_duplicates)
print("Diseases with <2 samples:", len(rare_classes))
print("Non-binary feature columns:", len(non_binary_cols))

```

Shape of dataset: (246945, 378)

First 5 rows:

	diseases	anxiety and nervousness	depression	shortness of breath
0	panic disorder	1	0	1
1	panic disorder	0	0	1
2	panic disorder	1	1	1
3	panic disorder	1	0	0
4	panic disorder	1	1	0

	depressive or psychotic symptoms	sharp chest pain	dizziness	insomnia
0	1	0	0	0
1	1	0	1	1
2	1	0	1	1
3	1	0	1	1
4	0	0	0	1

	abnormal involuntary movements	chest tightness	...	\
0	0	1	...	
1	0	0	...	
2	0	0	...	
3	1	0	...	
4	1	1	...	

	stuttering or stammering	problems with orgasm	nose deformity	\
0	0	0	0	
1	0	0	0	
2	0	0	0	
3	0	0	0	
4	0	0	0	

	lump over jaw	sore in nose	hip weakness	back swelling	\
0	0	0	0	0	
1	0	0	0	0	
2	0	0	0	0	
3	0	0	0	0	
4	0	0	0	0	

	ankle stiffness or tightness	ankle weakness	neck weakness
0	0	0	0
1	0	0	0
2	0	0	0
3	0	0	0
4	0	0	0

[5 rows x 378 columns]

Column names:

['diseases', 'anxiety and nervousness', 'depression', 'shortness of breath', 'depressive or psychotic symptoms', 'sharp chest pain', 'dizziness', 'insomnia', 'abnormal involuntary movements', 'chest tightness', 'palpitations', 'irregular heartbeat', 'breathing fast', 'hoarse voice', 'sore throat', 'difficulty speaking', 'cough', 'nasal congestion', 'throat swelling', 'diminished hearing', 'lump in throat', 'throat feels tight', 'difficulty in swallowin

g', 'skin swelling', 'retention of urine', 'groin mass', 'leg pain', 'hip pain', 'suprapubic pain', 'blood in stool', 'lack of growth', 'emotional symptoms', 'elbow weakness', 'back weakness', 'pus in sputum', 'symptoms of the scrotum and testes', 'swelling of scrotum', 'pain in testicles', 'flatulence', 'pus draining from ear', 'jaundice', 'mass in scrotum', 'white discharge from eye', 'irritable infant', 'abusing alcohol', 'fainting', 'hostile behavior', 'drug abuse', 'sharp abdominal pain', 'feeling ill', 'vomiting', 'headache', 'nausea', 'diarrhea', 'vaginal itching', 'vaginal dryness', 'painful urination', 'involuntary urination', 'pain during intercourse', 'frequent urination', 'lower abdominal pain', 'vaginal discharge', 'blood in urine', 'hot flashes', 'intermenstrual bleeding', 'hand or finger pain', 'wrist pain', 'hand or finger swelling', 'arm pain', 'wrist swelling', 'arm stiffness or tightness', 'arm swelling', 'hand or finger stiffness or tightness', 'wrist stiffness or tightness', 'lip swelling', 'toothache', 'abnormal appearing skin', 'skin lesion', 'acne or pimples', 'dry lips', 'facial pain', 'mouth ulcer', 'skin growth', 'eye deviation', 'diminished vision', 'double vision', 'cross-eyed', 'symptoms of eye', 'pain in eye', 'eye moves abnormally', 'abnormal movement of eyelid', 'foreign body sensation in eye', 'irregular appearing scalp', 'swollen lymph nodes', 'back pain', 'neck pain', 'low back pain', 'pain of the anus', 'pain during pregnancy', 'pelvic pain', 'impotence', 'infant spitting up', 'vomiting blood', 'regurgitation', 'burning abdominal pain', 'restlessness', 'symptoms of infants', 'wheezing', 'peripheral edema', 'neck mass', 'ear pain', 'jaw swelling', 'mouth dryness', 'neck swelling', 'knee pain', 'foot or toe pain', 'bowlegged or knock-kneed', 'ankle pain', 'bones are painful', 'knee weakness', 'elbow pain', 'knee swelling', 'skin moles', 'knee lump or mass', 'weight gain', 'problems with movement', 'knee stiffness or tightness', 'leg swelling', 'foot or toe swelling', 'heartburn', 'smoking problems', 'muscle pain', 'infant feeding problem', 'recent weight loss', 'problems with shape or size of breast', 'underweight', 'difficulty eating', 'scanty menstrual flow', 'vaginal pain', 'vaginal redness', 'vulvar irritation', 'weakness', 'decreased heart rate', 'increased heart rate', 'bleeding or discharge from nipple', 'ringing in ear', 'plugged feeling in ear', 'itchy ear(s)', 'frontal headache', 'fluid in ear', 'neck stiffness or tightness', 'spots or clouds in vision', 'eye redness', 'lacrimation', 'itchiness of eye', 'blindness', 'eye burns or stings', 'itchy eyelid', 'feeling cold', 'decreased appetite', 'excessive appetite', 'excessive anger', 'loss of sensation', 'focal weakness', 'slurring words', 'symptoms of the face', 'disturbance of memory', 'paresthesia', 'side pain', 'fever', 'shoulder pain', 'shoulder stiffness or tightness', 'shoulder weakness', 'arm cramps or spasms', 'shoulder swelling', 'tongue lesions', 'leg cramps or spasms', 'abnormal appearing tongue', 'ache all over', 'lower body pain', 'problems during pregnancy', 'spotting or bleeding during pregnancy', 'cramps and spasms', 'upper abdominal pain', 'stomach bloating', 'changes in stool appearance', 'unusual color or odor to urine', 'kidney mass', 'swollen abdomen', 'symptoms of prostate', 'leg stiffness or tightness', 'difficulty breathing', 'rib pain', 'joint pain', 'muscle stiffness or tightness', 'pallor', 'hand or finger lump or mass', 'chills', 'groin pain', 'fatigue', 'abdominal distention', 'regurgitation.1', 'symptoms of the kidneys', 'melena', 'flushing', 'coughing up sputum', 'seizures', 'delusions or hallucinations', 'shoulder cramps or spasms', 'joint stiffness or tightness', 'pain or soreness of breast', 'excessive urination at night', 'bleeding from eye', 'rectal bleeding', 'constipation', 'temper problems', 'coryza', 'wrist weakness', 'eye strain', 'hemoptysis', 'lymphedema', 'skin on leg or foot looks infected', 'allergic reaction', 'congestion in chest', 'muscle swelling', 'pus in urine', 'abnormal size or shape of ear', 'low back weakness', 'sleepiness', 'apnea', 'abnormal breathing sounds', 'excessive growth', 'elbow cramps or spasms', 'feeling hot and col

d', 'blood clots during menstrual periods', 'absence of menstruation', 'pulling at ears', 'gum pain', 'redness in ear', 'fluid retention', 'flu-like syndrome', 'sinus congestion', 'painful sinuses', 'fears and phobias', 'recent pregnancy', 'uterine contractions', 'burning chest pain', 'back cramps or spasms', 'stiffness all over', 'muscle cramps, contractures, or spasms', 'low back cramps or spasms', 'back mass or lump', 'nosebleed', 'long menstrual periods', 'heavy menstrual flow', 'unpredictable menstruation', 'painful menstruation', 'infertility', 'frequent menstruation', 'sweating', 'mass on eyelid', 'swollen eye', 'eyelid swelling', 'eyelid lesion or rash', 'unwanted hair', 'symptoms of bladder', 'irregular appearing nails', 'itching of skin', 'hurts to breath', 'nailbiting', 'skin dryness, peeling, scaliness, or roughness', 'skin on arm or hand looks infected', 'skin irritation', 'itchy scalp', 'hip swelling', 'incontinence of stool', 'foot or toe cramps or spasms', 'warts', 'bumps on penis', 'too little hair', 'foot or toe lump or mass', 'skin rash', 'mass or swelling around the anus', 'low back swelling', 'ankle swelling', 'hip lump or mass', 'drainage in throat', 'dry or flaky scalp', 'premenstrual tension or irritability', 'feeling hot', 'feet turned in', 'foot or toe stiffness or tightness', 'pelvic pressure', 'elbow swelling', 'elbow stiffness or tightness', 'early or late onset of menopause', 'mass on ear', 'bleeding from ear', 'hand or finger weakness', 'low self-esteem', 'throat irritation', 'itching of the anus', 'swollen or red tonsils', 'irregular belly button', 'swollen tongue', 'lip sore', 'vulvar sore', 'hip stiffness or tightness', 'mouth pain', 'arm weakness', 'leg lump or mass', 'disturbance of smell or taste', 'discharge in stools', 'penis pain', 'loss of sex drive', 'obsessions and compulsions', 'antisocial behavior', 'neck cramps or spasms', 'pupils unequal', 'poor circulation', 'thirst', 'sleepwalking', 'skin oiliness', 'sneezing', 'bladder mass', 'knee cramps or spasms', 'premature ejaculation', 'leg weakness', 'posture problems', 'bleeding in mouth', 'tongue bleeding', 'change in skin mole size or color', 'penis redness', 'penile discharge', 'shoulder lump or mass', 'polyuria', 'cloudy eye', 'hysterical behavior', 'arm lump or mass', 'nightmares', 'bleeding gums', 'pain in gums', 'bedwetting', 'diaper rash', 'lump or mass of breast', 'vaginal bleeding after menopause', 'infrequent menstruation', 'mass on vulva', 'jaw pain', 'itching of scrotum', 'postpartum problems of the breast', 'eyelid retracted', 'hesitancy', 'elbow lump or mass', 'muscle weakness', 'throat redness', 'joint swelling', 'tongue pain', 'redness in or around nose', 'wrinkles on skin', 'foot or toe weakness', 'hand or finger cramps or spasms', 'back stiffness or tightness', 'wrist lump or mass', 'skin pain', 'low back stiffness or tightness', 'low urine output', 'skin on head or neck looks infected', 'stuttering or stammering', 'problems with orgasm', 'nose deformity', 'lump over jaw', 'sore in nose', 'hip weakness', 'back swelling', 'ankle stiffness or tightness', 'ankle weakness', 'neck weakness']

Data types:

diseases	object
anxiety and nervousness	int64
depression	int64
shortness of breath	int64
depressive or psychotic symptoms	int64
	...
hip weakness	int64
back swelling	int64
ankle stiffness or tightness	int64
ankle weakness	int64
neck weakness	int64

Length: 378, dtype: object

Label column: diseases

Total missing values in dataset: 0

Columns with missing values:
Series([], dtype: int64)

Number of duplicate rows: 57298

Number of unique diseases: 773

Top 10 most common diseases:

diseases	
cystitis	1219
nose disorder	1218
vulvodynia	1218
complex regional pain syndrome	1217
spondylosis	1216
vaginal cyst	1215
esophagitis	1215
peripheral nerve disorder	1215
hypoglycemia	1215
conjunctivitis due to allergy	1215

Name: count, dtype: int64

Top 10 least common diseases:

diseases	
diabetes	1
thalassemia	1
heat stroke	1
gas gangrene	1
typhoid fever	1
open wound of the head	1
myocarditis	1
chronic ulcer	1
hypergammaglobulinemia	1
kaposi sarcoma	1

Name: count, dtype: int64

Summary of class counts:

count	773.000000
mean	319.463131
std	351.691549
min	1.000000
25%	30.000000
50%	168.000000
75%	505.000000
max	1219.000000

Name: count, dtype: float64

Number of feature columns: 377

Feature unique-value summary:

count	377.000000
mean	1.870027

```
std      0.336721
min      1.000000
25%      2.000000
50%      2.000000
75%      2.000000
max      2.000000
dtype: float64
```

Number of non-binary feature columns: 0

Top 20 most frequent symptoms:

sharp abdominal pain	32307
vomiting	27874
headache	24719
cough	24296
sharp chest pain	24016
nausea	23687
back pain	21809
shortness of breath	21346
fever	20394
dizziness	17272
abnormal appearing skin	16425
nasal congestion	16200
leg pain	16189
skin swelling	15165
depressive or psychotic symptoms	15064
lower abdominal pain	14936
sore throat	14005
burning abdominal pain	12981
skin rash	12375
skin lesion	12340

dtype: int64

Top 20 least frequent symptoms:

elbow stiffness or tightness	0
mass on ear	0
throat irritation	0
swollen tongue	0
sleepwalking	0
bleeding in mouth	0
knee cramps or spasms	0
posture problems	0
low back stiffness or tightness	0
skin oiliness	0
skin on head or neck looks infected	0
stuttering or stammering	0
polyuria	0
tongue bleeding	0
change in skin mole size or color	0
lump over jaw	0
hip weakness	0
back swelling	0
ankle stiffness or tightness	0
neck weakness	0

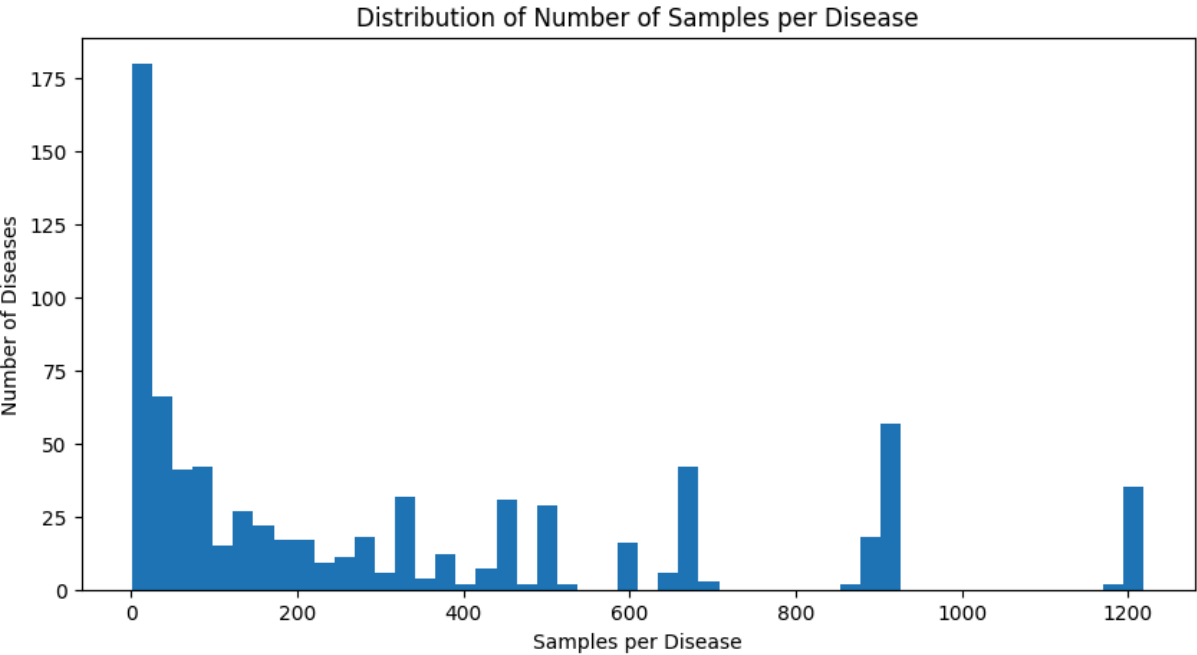
dtype: int64

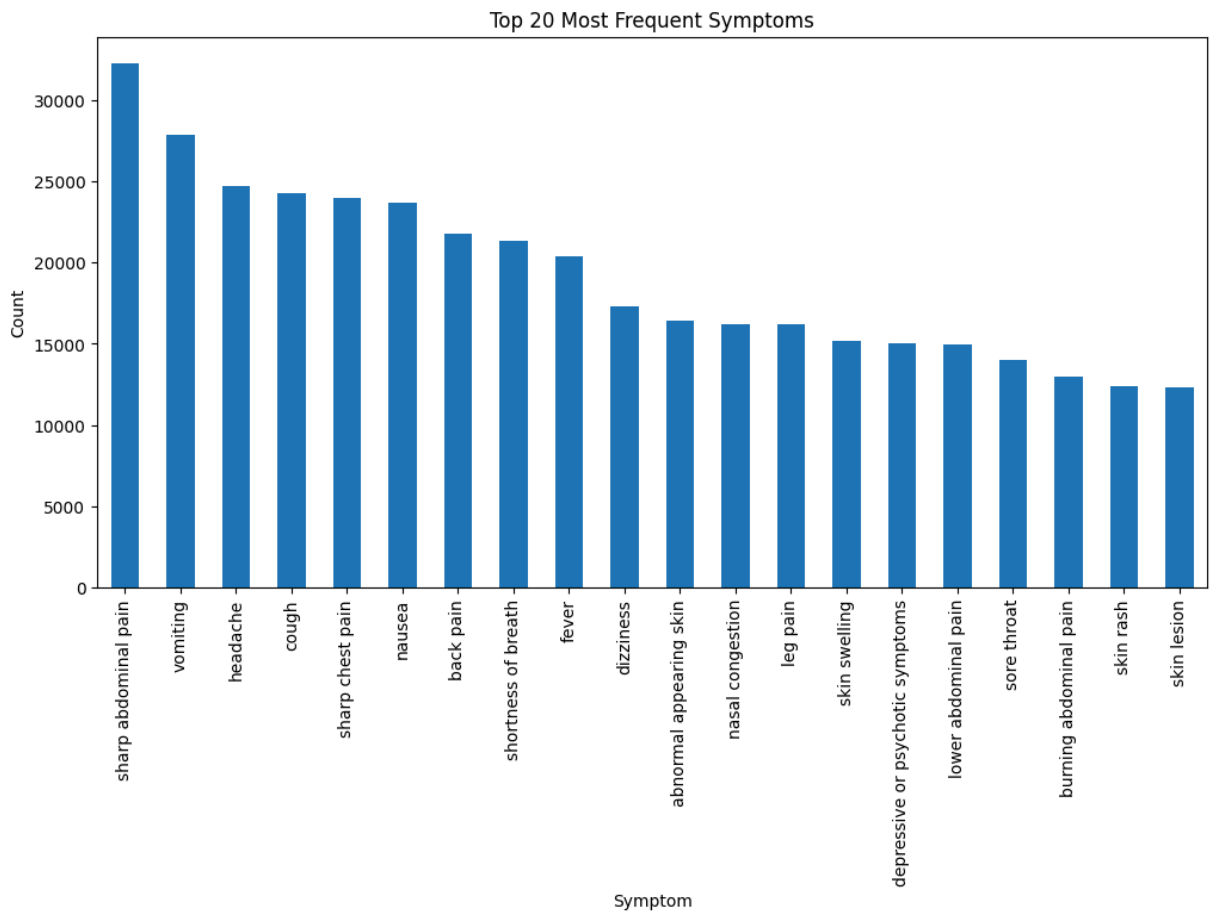
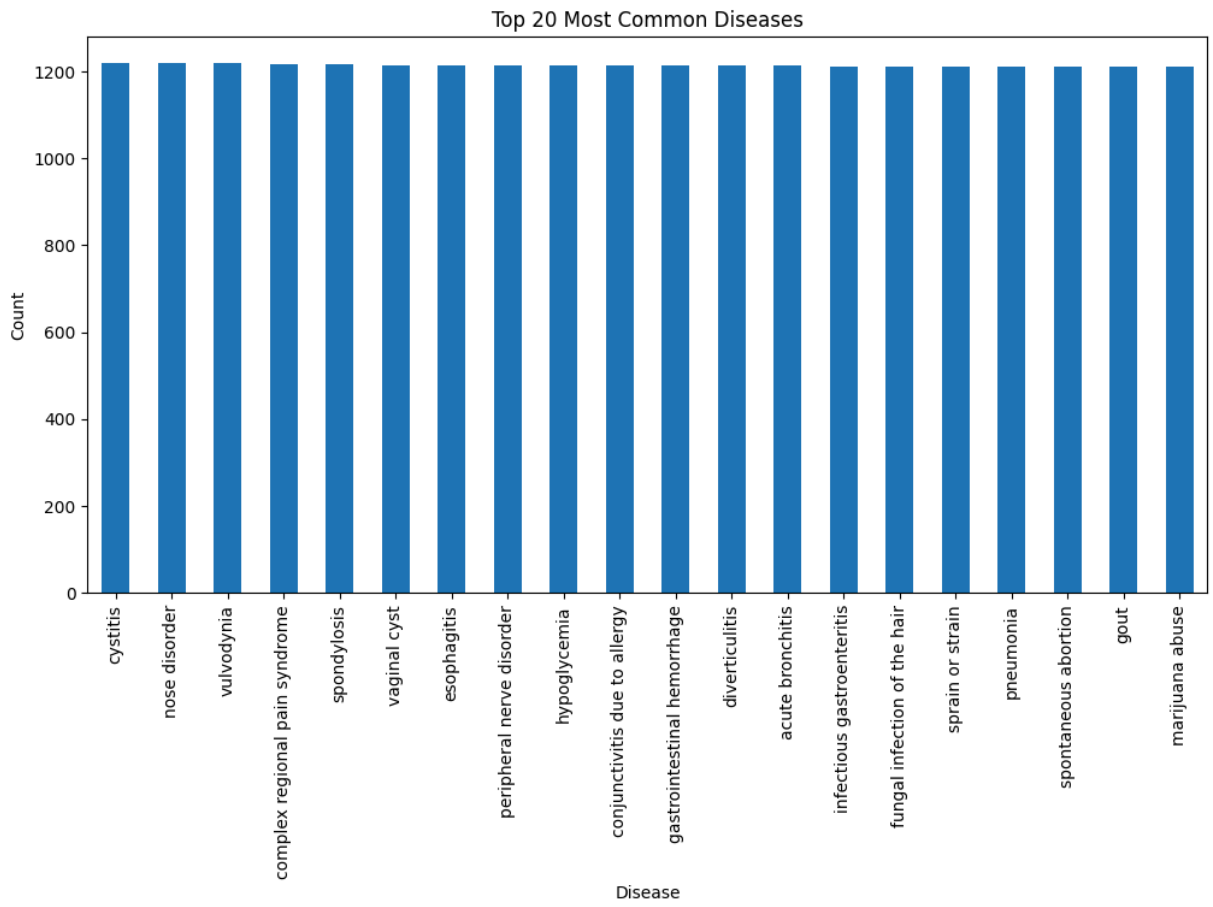
Number of diseases with fewer than 2 samples: 19

Diseases with fewer than 2 samples:

diseases	
turner syndrome	1
rocky mountain spotted fever	1
open wound of the cheek	1
high blood pressure	1
open wound due to trauma	1
open wound of the chest	1
huntington disease	1
open wound of the knee	1
foreign body in the nose	1
diabetes	1
thalassemia	1
heat stroke	1
gas gangrene	1
typhoid fever	1
open wound of the head	1
myocarditis	1
chronic ulcer	1
hypergammaglobulinemia	1
kaposi sarcoma	1

Name: count, dtype: int64





Final EDA Summary

Rows: 246945
Columns: 378
Features: 377
Unique diseases: 773
Total missing values: 0
Duplicate rows: 57298
Diseases with <2 samples: 19
Non-binary feature columns: 0

```
In [ ]: # Drop rows where label is missing
df_clean = df.dropna(subset=[label_col]).copy()

# Fill missing feature values with 0 (symptom absent)
df_clean = df_clean.fillna(0)

# Remove duplicates
df_clean = df_clean.drop_duplicates()

print("Final shape:", df_clean.shape)
```

Final shape: (189647, 378)

```
In [ ]: from sklearn.preprocessing import LabelEncoder
y = df_clean[label_col]
class_counts = y.value_counts()
# Remove rare classes from dataset
valid_classes = class_counts[class_counts >= 10].index

df_clean = df_clean[df_clean[label_col].isin(valid_classes)].copy()

df_clean = df_clean.sample(n=100000, random_state=42)
```

```
In [ ]: # redo X, y after filtering
X = df_clean.drop(columns=[label_col])
y = df_clean[label_col]
y_encoded = LabelEncoder().fit_transform(y)
```

```
In [ ]:
```

Kmean

```
In [ ]: #reduce with PCA
from sklearn.decomposition import PCA

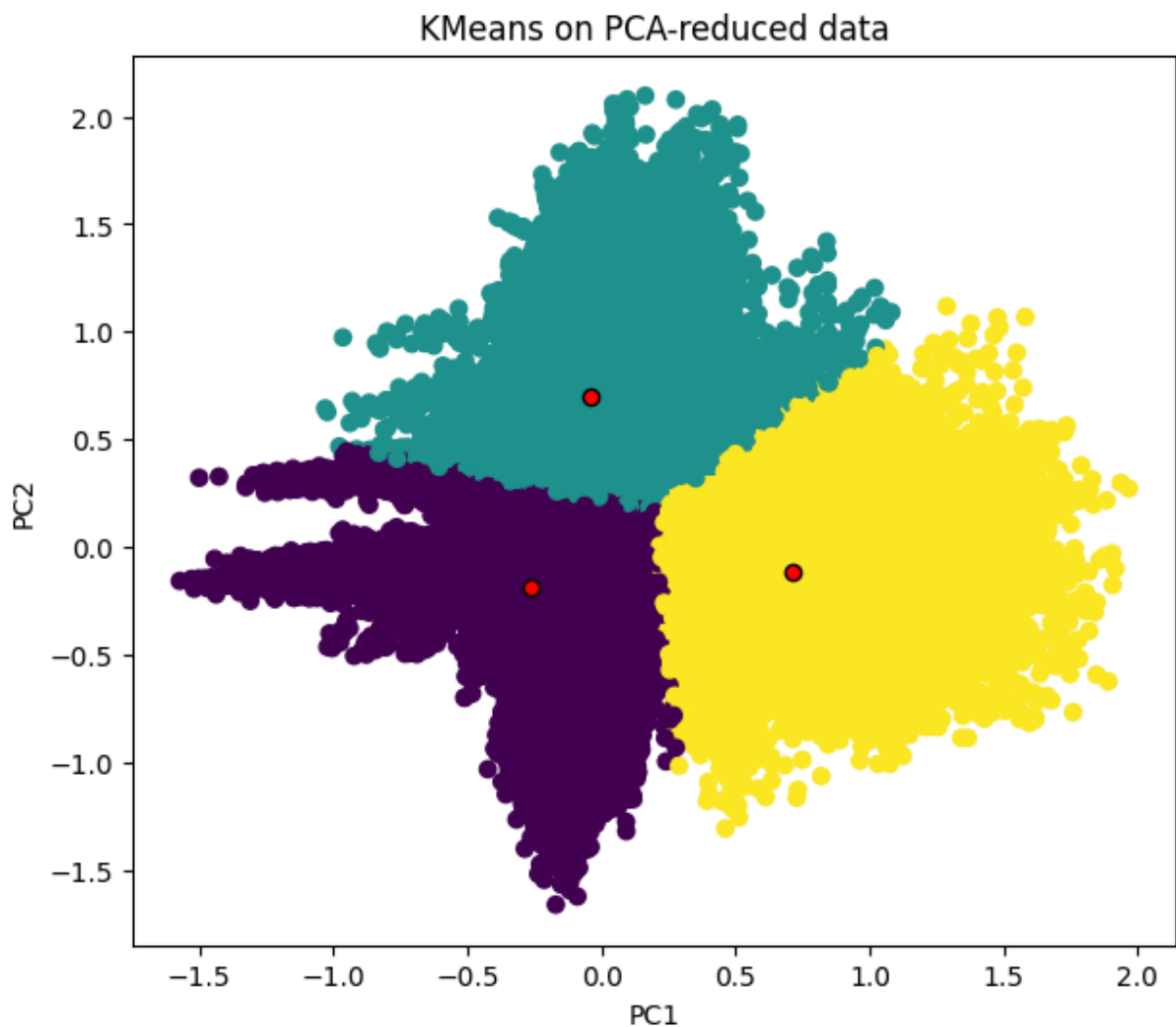
X = df_clean.drop(columns=[label_col])

pca = PCA(n_components=2)
X_pca = pca.fit_transform(X)
```

```
In [ ]: from sklearn.cluster import KMeans

#kmeans clustering
k = 3
kmeans = KMeans(n_clusters=k, random_state=0)
y_predicted = kmeans.fit_predict(X_pca)
```

```
In [ ]: import matplotlib.pyplot as plt
centroids = kmeans.cluster_centers_
plt.figure(figsize=(7,6))
plt.scatter(X_pca[:, 0], X_pca[:, 1], c=y_predicted)
plt.scatter(centroids[:, 0], centroids[:, 1],
            c='red',
            edgecolor='black', # Outline for visibility
            label='Centroids')
plt.xlabel("PC1")
plt.ylabel("PC2")
plt.title("KMeans on PCA-reduced data")
plt.show()
```



```
In [ ]: import pandas as pd
import numpy as np
```

```

# 1. Get the centroids in PCA space
centroids_pca = kmeans.cluster_centers_

# 2. Inverse transform them back to the original feature space
# This gives us a 'profile' of the average symptom values for that cluster
centroids_original = pca.inverse_transform(centroids_pca)

# 3. Create a DataFrame for better readability
feature_names = X.columns
df_centroids = pd.DataFrame(centroids_original, columns=feature_names)

print("---- Characteristics of the 3 Cluster Centroids ----")
for i in range(k):
    print(f"\nCluster {i} Top Symptoms:")
    # Sort symptoms by their 'intensity' or frequency in the centroid
    top_symptoms = df_centroids.iloc[i].sort_values(ascending=False).head(5)
    print(top_symptoms)

# Sum the values for each cluster (row) across all symptom columns
avg_symptom_counts = df_centroids.sum(axis=1)

print("---- Average Number of Symptoms per Cluster ----")
for i, count in enumerate(avg_symptom_counts):
    print(f"Cluster {i}: {count:.2f} symptoms")

```

---- Characteristics of the 3 Cluster Centroids ----

Cluster 0 Top Symptoms:

leg pain	0.120197
abnormal appearing skin	0.119246
back pain	0.115483
skin swelling	0.111499
skin lesion	0.097975

Name: 0, dtype: float64

Cluster 1 Top Symptoms:

cough	0.379764
fever	0.315551
nasal congestion	0.277149
sore throat	0.230537
shortness of breath	0.224890

Name: 1, dtype: float64

Cluster 2 Top Symptoms:

sharp abdominal pain	0.485616
nausea	0.345935
vomiting	0.344935
back pain	0.235236
burning abdominal pain	0.210266

Name: 2, dtype: float64

---- Average Number of Symptoms per Cluster ----

Cluster 0: 5.29 symptoms

Cluster 1: 5.48 symptoms

Cluster 2: 5.70 symptoms

```
In [ ]: # Print the coordinates to match them with the plot
        for i, center in enumerate(kmeans.cluster_centers_):
            print(f"Cluster {i} (from your 'Top Symptoms' list) is at PCA coordinate
```

Cluster 0 (from your 'Top Symptoms' list) is at PCA coordinates: [-0.2639408
6 -0.18921203]

Cluster 1 (from your 'Top Symptoms' list) is at PCA coordinates: [-0.0405248
2 0.69963048]

Cluster 2 (from your 'Top Symptoms' list) is at PCA coordinates: [0.7117142
9 -0.11775276]

```
In [ ]: from sklearn.cluster import KMeans
        from sklearn.metrics import silhouette_score

        scores = []

        for k in range(2, 20):
            kmeans = KMeans(n_clusters=k, random_state=0)
            labels = kmeans.fit_predict(X_pca)
            score = silhouette_score(X_pca, labels)
            scores.append(score)
            print(f"k={k}, score={score}")
```

k=2, score=0.374754938990626
k=3, score=0.45454067311513424
k=4, score=0.3757302745174936
k=5, score=0.43059354624115853
k=6, score=0.4245918544583044
k=7, score=0.35508798510723194
k=8, score=0.4234403032902573
k=9, score=0.3827407053728113
k=10, score=0.40844456812320423
k=11, score=0.4061677250070966
k=12, score=0.37246582150286706
k=13, score=0.3632117677527574
k=14, score=0.35930205767142886
k=15, score=0.368650412152064
k=16, score=0.3760732830897656
k=17, score=0.3618924009668253
k=18, score=0.3615548848270993
k=19, score=0.36372130024122923

```
In [ ]: from sklearn.cluster import KMeans
        from sklearn.metrics import silhouette_score

        scores = []

        for k in range(700, 700):
            kmeans = KMeans(n_clusters=k, random_state=0)
            labels = kmeans.fit_predict(X_pca)
            score = silhouette_score(X_pca, labels)
            scores.append(score)
            print(f"k={k}, score={score}")
```

```

-----
AttributeError                                Traceback (most recent call last)
/tmp/ipykernel_1351/2445306000.py in <cell line: 0>()
----> 1 from sklearn.cluster import KMeans
      2 from sklearn.metrics import silhouette_score
      3
      4 scores = []
      5

/usr/local/lib/python3.12/dist-packages/sklearn/__init__.py in <module>
      71     _distributor_init,
      72 )
----> 73 from .base import clone # noqa: E402
      74 from .utils._show_versions import show_versions # noqa: E402
      75

/usr/local/lib/python3.12/dist-packages/sklearn/base.py in <module>
      17 from ._config import config_context, get_config
      18 from .exceptions import InconsistentVersionWarning
----> 19 from .utils._estimator_html_repr import _HTMLDocumentationLinkMixin,
estimator_html_repr
      20 from .utils._metadata_requests import _MetadataRequester, _routing_e
nabled
      21 from .utils._param_validation import validate_parameter_constraints

/usr/local/lib/python3.12/dist-packages/sklearn/utils/__init__.py in <module>
>
      13 from . import _joblib, metadata_routing
      14 from ._bunch import Bunch
----> 15 from ._chunking import gen_batches, gen_even_slices
      16 from ._estimator_html_repr import estimator_html_repr
      17

/usr/local/lib/python3.12/dist-packages/sklearn/utils/_chunking.py in <modul
e>
      9
      10 from .._config import get_config
----> 11 from ._param_validation import Interval, validate_params
      12
      13

/usr/local/lib/python3.12/dist-packages/sklearn/utils/_param_validation.py i
n <module>
      15
      16 from .._config import config_context, get_config
----> 17 from .validation import _is_arraylike_not_scalar
      18
      19

/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py in <modu
le>
      19 from .. import get_config as _get_config
      20 from ..exceptions import DataConversionWarning, NotFittedError, Posi
tiveSpectrumWarning
----> 21 from ..utils._array_api import _asarray_with_order, _is_numpy_namesp
ace, get_namespace

```

```

22 from ..utils.deprecation import _deprecate_force_all_finite
23 from ..utils.fixes import ComplexWarning, _preserve_dia_indices_dtype
e

/usr/local/lib/python3.12/dist-packages/sklearn/utils/_array_api.py in <module>
15
16 from .._config import get_config
----> 17 from .fixes import parse_version
18
19 _NUMPY_NAMESPACE_NAMES = {"numpy", "array_api_compat.numpy"}

/usr/local/lib/python3.12/dist-packages/sklearn/utils/fixes.py in <module>
18
19 try:
----> 20     import pandas as pd
21 except ImportError:
22     pd = None

/usr/local/lib/python3.12/dist-packages/pandas/__init__.py in <module>
137
138 from pandas import api, arrays, errors, io, plotting, tseries
--> 139 from pandas import testing
140 from pandas.util._print_versions import show_versions
141

/usr/local/lib/python3.12/dist-packages/pandas/testing.py in <module>
4
5
----> 6 from pandas._testing import (
7     assert_extension_array_equal,
8     assert_frame_equal,

/usr/local/lib/python3.12/dist-packages/pandas/_testing/__init__.py in <module>
le>
402
403
--> 404 cython_table = pd.core.common._cython_table.items()
405
406

AttributeError: partially initialized module 'pandas' has no attribute 'core'
e' (most likely due to a circular import)

```

```

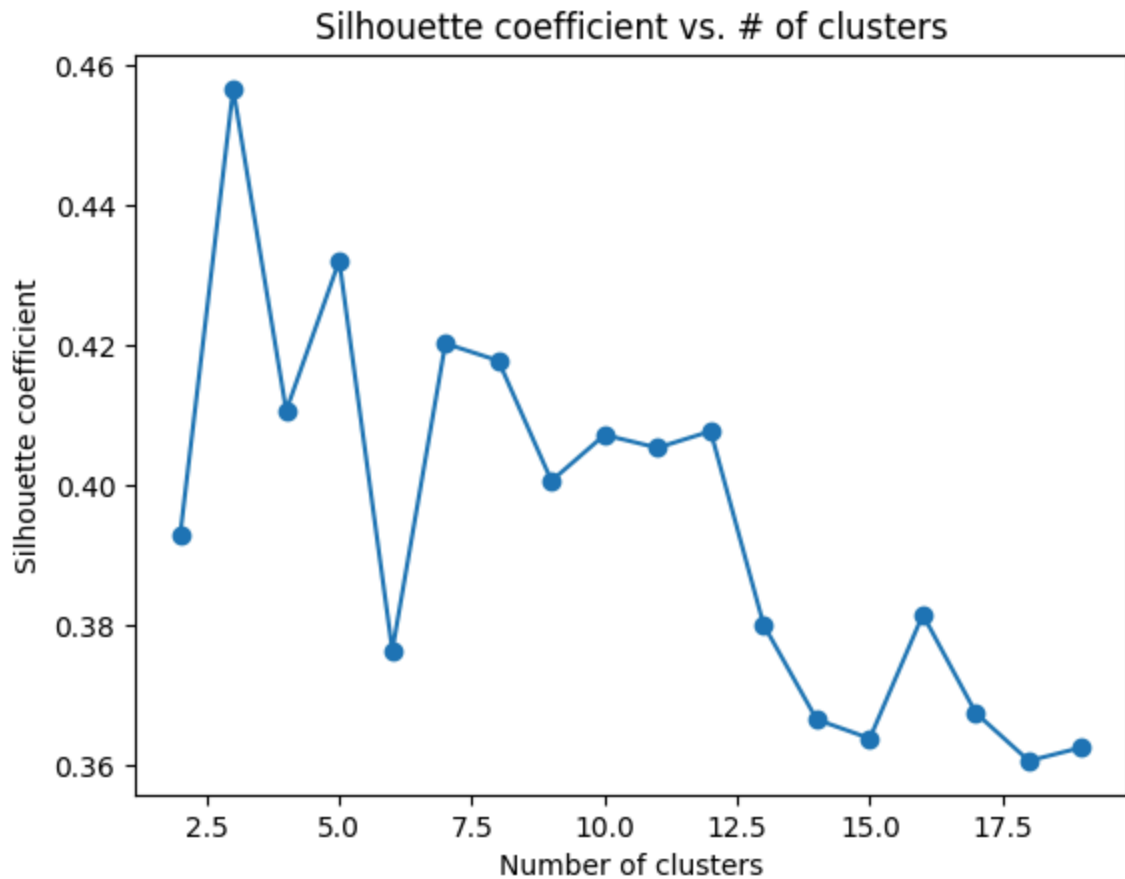
In [ ]: plt.scatter(range(2, 20), scores)
plt.plot(range(2, 20), scores)
plt.xlabel("Number of clusters")
plt.ylabel("Silhouette coefficient")
plt.title("Silhouette coefficient vs. # of clusters")

```

```

Out[ ]: Text(0.5, 1.0, 'Silhouette coefficient vs. # of clusters')

```



In []:

Language model

In []: `%pip install --upgrade --quiet google-genai gitingest`

```
_____ 52.4/52.4 kB 2.3 MB/s eta 0:00
0:00
_____ 786.1/786.1 kB 12.6 MB/s eta 0:00
0:00
_____ 68.3/68.3 kB 2.8 MB/s eta 0:00:00
0
_____ 240.6/240.6 kB 17.8 MB/s eta 0:00
0:00
_____ 61.6/61.6 kB 4.2 MB/s eta 0:00:00
0
_____ 55.2/55.2 kB 3.8 MB/s eta 0:00:00
0
```

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.
google-colab 1.0.0 requires google-auth==2.47.0, but you have google-auth 2.49.2 which is incompatible.

In []: `import sys`

```
# Additional authentication is required for Google Colab
if "google.colab" in sys.modules:
    # Authenticate user to Google Cloud
    from google.colab import auth

    auth.authenticate_user()
```

```
In [ ]: import os

from google import genai

# fmt: off
PROJECT_ID = "dsci303-test" # @param {type: "string", placeholder: "[your-p

# fmt: on
if not PROJECT_ID or PROJECT_ID == "[your-project-id]":
    PROJECT_ID = str(os.environ.get("GOOGLE_CLOUD_PROJECT"))

LOCATION = os.environ.get("GOOGLE_CLOUD_REGION", "global")

client = genai.Client(vertexai=True, project=PROJECT_ID, location=LOCATION)
```

```
In [ ]: import nest_asyncio
from IPython.display import Audio, Image, Markdown, Video, display
from gitingest import ingest
from google.genai.types import CreateCachedContentConfig, GenerateContentCor

nest_asyncio.apply()
```

```
In [ ]: MODEL_ID = "gemini-2.5-flash" # @param {type: "string"}
```

```
In [ ]: import random
import json

# number of samples
N = 8

# sample rows
df_sample = df_clean.sample(n=N, random_state=0)

X_sample = df_sample.drop(columns=[label_col])
y_sample = df_sample[label_col].astype(str)

allowed_labels = sorted(df_clean[label_col].astype(str).unique())

def row_to_text(row):
    active = []
    for col, val in row.items():
        if val == 1:
            active.append(col)
        elif val not in (0, 1):
            active.append(f"{col}={val}")
    return ", ".join(active) if active else "No symptoms"

def build_prompt(symptoms):
```

```

        return f"""
Choose exactly ONE disease from this list:
{", ".join(allowed_labels)}

Symptoms:
{symptoms}

Return JSON:
{{"predicted_disease": "...", "reasoning": "..."}}
"""

correct = 0

for i, (idx, row) in enumerate(X_sample.iterrows()):
    symptoms = row_to_text(row)
    true_label = y_sample.loc[idx]

    prompt = build_prompt(symptoms)

    response = client.models.generate_content(
        model="gemini-3.1-pro-preview", # or your model
        contents=prompt
    )

    text = response.text.strip()

    try:
        data = json.loads(text)
        pred = data.get("predicted_disease", "").strip()
        reasoning = data.get("reasoning", "")
    except:
        pred = "PARSE_ERROR"
        reasoning = text

    is_correct = (pred == true_label)
    if is_correct:
        correct += 1

    print(f"\nExample {i+1}")
    print("Symptoms:", symptoms[:200])
    print("True:", true_label)
    print("Pred:", pred)
    print("Correct:", is_correct)
    print("Reasoning:", reasoning[:150])

# final accuracy
print("\n=====")
print(f"Correct: {correct}/{N}")
print(f"Accuracy: {correct/N:.2f}")

```

```
2026-04-14 04:08:45.481 | INFO      | logging:callHandlers:1762 | AFC is enabled with max remote calls: 10.
2026-04-14 04:09:00.056 | INFO      | logging:callHandlers:1762 | HTTP Request: POST https://aiplatform.googleapis.com/v1beta1/projects/dsci303-test/locations/global/publishers/google/models/gemini-3.1-pro-preview:generateContent "HTTP/1.1 200 OK"
2026-04-14 04:09:00.063 | INFO      | logging:callHandlers:1762 | AFC is enabled with max remote calls: 10.
```

Example 1

Symptoms: shortness of breath, nasal congestion, vomiting, weakness, fever, coryza

True: pneumonia

Pred: PARSE_ERROR

Correct: False

Reasoning: ```json

```
{
  "predicted_disease": "acute bronchiolitis",
  "reasoning": "The combination of upper respiratory symptoms (coryza, nasal congestion) with
```

```
2026-04-14 04:09:24.937 | INFO      | logging:callHandlers:1762 | HTTP Request: POST https://aiplatform.googleapis.com/v1beta1/projects/dsci303-test/locations/global/publishers/google/models/gemini-3.1-pro-preview:generateContent "HTTP/1.1 200 OK"
```

```
2026-04-14 04:09:24.942 | INFO      | logging:callHandlers:1762 | AFC is enabled with max remote calls: 10.
```

Example 2

Symptoms: anxiety and nervousness, depression, depressive or psychotic symptoms, lack of growth, fainting, hostile behavior, antisocial behavior

True: asperger syndrome

Pred: PARSE_ERROR

Correct: False

Reasoning: ```json

```
{
  "predicted_disease": "eating disorder",
  "reasoning": "The combination of symptoms strongly suggests a condition with both severe psychia
```

```
2026-04-14 04:09:37.717 | INFO      | logging:callHandlers:1762 | HTTP Request: POST https://aiplatform.googleapis.com/v1beta1/projects/dsci303-test/locations/global/publishers/google/models/gemini-3.1-pro-preview:generateContent "HTTP/1.1 200 OK"
```

```
2026-04-14 04:09:37.722 | INFO      | logging:callHandlers:1762 | AFC is enabled with max remote calls: 10.
```

Example 3

Symptoms: anxiety and nervousness, depression, depressive or psychotic symptoms, insomnia, excessive anger, delusions or hallucinations, temper problems

True: impulse control disorder

Pred: PARSE_ERROR

Correct: False

Reasoning: ```json

```
{
  "predicted_disease": "bipolar disorder",
  "reasoning": "The combination of depressive symptoms, insomnia, and anxiety, along with excessi
```

2026-04-14 04:09:53.213 | INFO | logging:callHandlers:1762 | HTTP Request: POST https://aiplatform.googleapis.com/v1beta1/projects/dsci303-test/locations/global/publishers/google/models/gemini-3.1-pro-preview:generateContent "HTTP/1.1 200 OK"

2026-04-14 04:09:53.218 | INFO | logging:callHandlers:1762 | AFC is enabled with max remote calls: 10.

Example 4

Symptoms: dizziness, chest tightness, palpitations, fainting, feeling ill, weakness

True: sinus bradycardia

Pred: PARSE_ERROR

Correct: False

Reasoning: ```json

```
{
  "predicted_disease": "paroxysmal supraventricular tachycardia",
  "reasoning": "The combination of sudden palpitations, dizziness, chest t
```

2026-04-14 04:11:21.867 | INFO | logging:callHandlers:1762 | HTTP Request: POST https://aiplatform.googleapis.com/v1beta1/projects/dsci303-test/locations/global/publishers/google/models/gemini-3.1-pro-preview:generateContent "HTTP/1.1 200 OK"

2026-04-14 04:11:21.876 | INFO | logging:callHandlers:1762 | AFC is enabled with max remote calls: 10.

Example 5

Symptoms: difficulty speaking, headache, double vision, back pain, disturbance of memory, rib pain, sleepiness

True: concussion

Pred: PARSE_ERROR

Correct: False

Reasoning: ```json

```
{
  "predicted_disease": "metastatic cancer",
  "reasoning": "The patient's symptoms can be distinctly grouped into two categories: focal and
```

2026-04-14 04:11:31.332 | INFO | logging:callHandlers:1762 | HTTP Request: POST https://aiplatform.googleapis.com/v1beta1/projects/dsci303-test/locations/global/publishers/google/models/gemini-3.1-pro-preview:generateContent "HTTP/1.1 200 OK"

2026-04-14 04:11:31.337 | INFO | logging:callHandlers:1762 | AFC is enabled with max remote calls: 10.

Example 6

Symptoms: chest tightness, cough, fever

True: asthma

Pred: PARSE_ERROR

Correct: False

Reasoning: ```json

```
{
  "predicted_disease": "acute bronchitis",
  "reasoning": "The combination of chest tightness, cough, and fever strongly suggests an acute l
```

2026-04-14 04:11:46.703 | INFO | logging:callHandlers:1762 | HTTP Request: POST https://aiplatform.googleapis.com/v1beta1/projects/dsci303-test/locations/global/publishers/google/models/gemini-3.1-pro-preview:generateContent "HTTP/1.1 200 OK"

2026-04-14 04:11:46.716 | INFO | logging:callHandlers:1762 | AFC is enabled with max remote calls: 10.

Example 7

Symptoms: sore throat, irritable infant, mouth ulcer, fever, constipation, skin rash

True: oral thrush (yeast infection)

Pred: PARSE_ERROR

Correct: False

Reasoning: ```json

{

"predicted_disease": "herpangina",

"reasoning": "The symptoms of sore throat, mouth ulcers, fever, and an irritable infant point strongly

2026-04-14 04:12:15.475 | INFO | logging:callHandlers:1762 | HTTP Request: POST https://aiplatform.googleapis.com/v1beta1/projects/dsci303-test/locations/global/publishers/google/models/gemini-3.1-pro-preview:generateContent "HTTP/1.1 200 OK"

Example 8

Symptoms: leg pain, side pain, stiffness all over, back mass or lump

True: spondylitis

Pred: PARSE_ERROR

Correct: False

Reasoning: ```json

{

"predicted_disease": "kidney cancer",

"reasoning": "The combination of 'side pain' (flank pain) and a 'back mass or lump' (palpable flank

=====

Correct: 0/8

Accuracy: 0.00

In []: